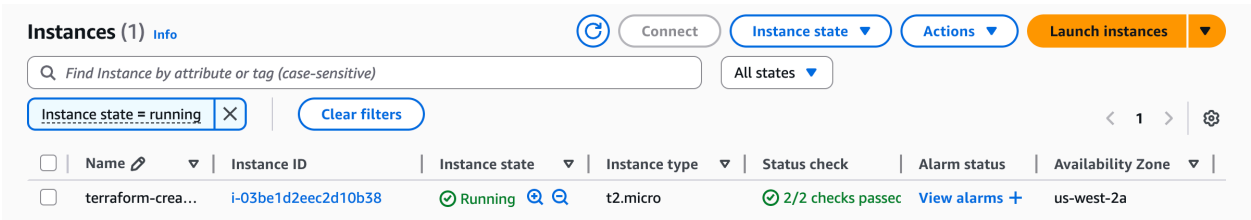
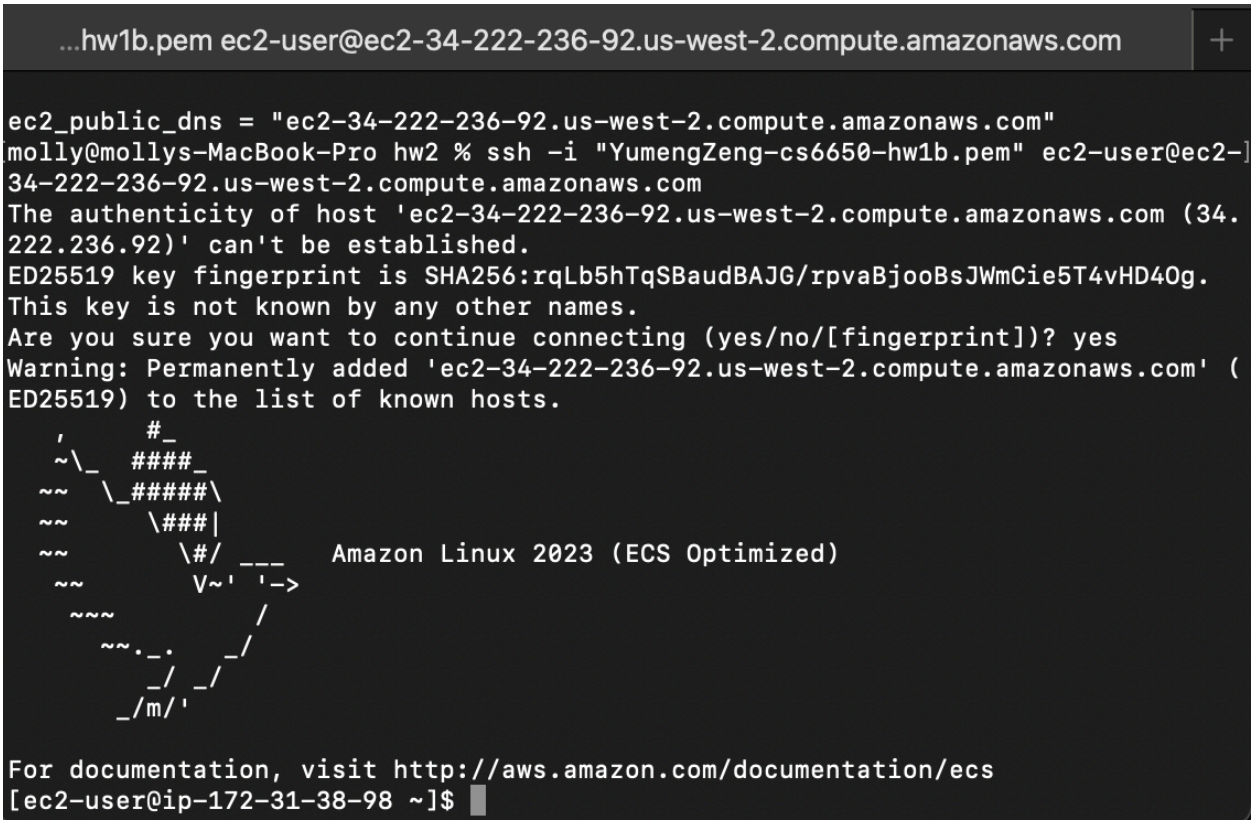
HW2_Screenshots

1. PART2

part2_terraform_apply.png



part2_ssh_success.png



Victory: [GIN-debug] Listening and serving HTTP on: 8080

```
molly@mollys-MacBook-Pro hw2 % docker image ls
                                                    i Info →   U  In Use
IMAGE               ID             DISK USAGE   CONTENT SIZE   EXTRA
my-go-app:latest    951f20a38414        920MB          190MB
molly@mollys-MacBook-Pro hw2 % docker run -d -p 8080:8080 --name my-running-app my-go-app
b31301de08106121b87ea9dd27dfa9042e36cbd8343916dd39c7d87886b8f7dc
molly@mollys-MacBook-Pro hw2 % docker logs my-running-app
[GIN-debug] [WARNING] Creating an Engine instance with the Logger and Recovery middleware already attached.

[GIN-debug] [WARNING] Running in "debug" mode. Switch to "release" mode in production.
 - using env:   export GIN_MODE=release
 - using code:  gin.SetMode(gin.ReleaseMode)

[GIN-debug] GET    /albums                   --> main.getAlbums (3 handlers)
[GIN-debug] GET    /albums/:id               --> main.getAlbumByID (3 handlers)
[GIN-debug] POST   /albums                   --> main.postAlbums (3 handlers)
[GIN-debug] [WARNING] You trusted all proxies, this is NOT safe. We recommend you to set a value.
Please check https://github.com/gin-gonic/gin/blob/master/docs/doc.md#dont-trust-all-proxies for details.
[GIN-debug] Listening and serving HTTP on :8080
```

Victory: http://localhost:8080/albums

```
molly@mollys-MacBook-Pro hw2 % curl http://localhost:8080/albums
[
    {
        "id": "1",
        "title": "Blue Train",
        "artist": "John Coltrane",
        "price": 56.99
    },
    {
        "id": "2",
        "title": "Jeru",
        "artist": "Gerry Mulligan",
        "price": 17.99
    },
    {
        "id": "3",
        "title": "Sarah Vaughan and Clifford Brown",
        "artist": "Sarah Vaughan",
        "price": 39.99
    }
```

2. PART3

Victory: curl http://localhost:8080/albums

```
[ec2-user@ip-172-31-38-98 ~]$ cd cs6650-distributed-systems/hw2
[ec2-user@ip-172-31-38-98 hw2]$ docker build -t my-go-app .
[+] Building 321.2s (14/14) FINISHED                              docker:default
 => [internal] load build definition from Dockerfile                          0.0s
 => => transferring dockerfile: 743B                                          0.0s
 => resolve image config for docker.io/docker/dockerfile:1                    0.7s
 => CACHED docker-image://docker.io/docker/dockerfile:1@sha256:b6afd42430     0.0s
 => [internal] load build definition from Dockerfile                          0.0s
 => [internal] load metadata for docker.io/library/golang:1.25-alpine         0.5s
 => [internal] load .dockerignore                                             0.0s
 => => transferring context: 2B                                               0.0s
 => [1/6] FROM docker.io/library/golang:1.25-alpine@sha256:d9b2e14101f27e     0.0s
 => [internal] load build context                                             0.0s
 => => transferring context: 258B                                             0.0s
 => CACHED [2/6] WORKDIR /app                                                 0.0s
 => CACHED [3/6] COPY go.mod go.sum ./                                        0.0s
 => CACHED [4/6] RUN go mod download                                          0.0s
 => CACHED [5/6] COPY *.go ./                                                 0.0s
 => [6/6] RUN CGO_ENABLED=0 GOOS=linux go build -o /godocker                306.2s
 => exporting to image                                                       12.9s
 => => exporting layers                                                      12.9s
 => => writing image sha256:c24d843cd6ec0d835227497bcdb810d1caa5b4e63827f     0.0s
 => => naming to docker.io/library/my-go-app                                  0.0s
[ec2-user@ip-172-31-38-98 hw2]$ docker run -d -p 8080:8080 --name my-running-app my-go-app
303a7d35123a9ab55cf52445100c4eeff652fd9a07ad1789599b454c749cd535
[ec2-user@ip-172-31-38-98 hw2]$ curl http://localhost:8080/albums
[
    {
        "id": "1",
        "title": "Blue Train",
        "artist": "John Coltrane",
        "price": 56.99
    },
    {
        "id": "2",
        "title": "Jeru",
        "artist": "Gerry Mulligan",
        "price": 17.99
```

```
[ec2-user@ip-172-31-38-98 hw2]$ docker images
REPOSITORY                  TAG            IMAGE ID       CREATED        SIZE
my-go-app                   latest         c24d843cd6ec   12 hours ago   641MB
amazon/amazon-ecs-agent     latest         192298286f1f   13 days ago    120MB
ebs-csi-driver              latest         702835b18362   13 days ago    58.8MB
ecs-service-connect-agent   interface-v1   369a1f77771a   4 weeks ago    204MB
amazon/amazon-ecs-pause     0.1.0          9dd4685d3644   11 years ago   702kB
[ec2-user@ip-172-31-38-98 hw2]$ docker rm -f my-running-app
my-running-app
[ec2-user@ip-172-31-38-98 hw2]$ ls
Dockerfile  go.mod  go.sum  main.go  main.tf  terraform.tfvars
[ec2-user@ip-172-31-38-98 hw2]$ docker run -d -p 8080:8080 --name my-running-app my-go-app
c46207b31c648618782f5ca241f16f855365024f288c66a0c42925066fbea7f6
[ec2-user@ip-172-31-38-98 hw2]$ curl http://localhost:8080/albums
[
    {
        "id": "1",
        "title": "Blue Train",
        "artist": "John Coltrane",
        "price": 56.99
    },
    {
        "id": "2",
        "title": "Jeru",
        "artist": "Gerry Mulligan",
        "price": 17.99
    },
    {
        "id": "3",
        "title": "Sarah Vaughan and Clifford Brown",
        "artist": "Sarah Vaughan",
        "price": 39.99
    }
][ec2-user@ip-172-31-38-98 hw2]$
```

Victory: curl http://16.145.86.114:8080/albums (From Public IPv4 address)

```
[ec2-user@ip-172-31-38-98 hw2]$ curl http://16.145.86.114:8080/albums
[
    {
        "id": "1",
        "title": "Blue Train",
        "artist": "John Coltrane",
        "price": 56.99
    },
    {
        "id": "2",
        "title": "Jeru",
        "artist": "Gerry Mulligan",
        "price": 17.99
    },
    {
        "id": "3",
        "title": "Sarah Vaughan and Clifford Brown",
        "artist": "Sarah Vaughan",
        "price": 39.99
    }
```

Not Secure 16.145.86.114:8080/albums

Pretty-print


```
[
    {
        "id": "1",
        "title": "Blue Train",
        "artist": "John Coltrane",
        "price": 56.99
    },
    {
        "id": "2",
        "title": "Jeru",
        "artist": "Gerry Mulligan",
        "price": 17.99
    },
    {
        "id": "3",
        "title": "Sarah Vaughan and Clifford Brown",
        "artist": "Sarah Vaughan",
        "price": 39.99
    }
]
```


3. PART4

Running two instances:

```
hw2 — -zsh — 80×24
aws_instance.demo-instance[0]: Modifying... [id=i-03be1d2eec2d10b38]
aws_instance.demo-instance[0]: Modifications complete after 2s [id=i-03be1d2eec2
d10b38]
aws_instance.demo-instance[1]: Still creating... [00m10s elapsed]
aws_instance.demo-instance[1]: Still creating... [00m20s elapsed]
aws_instance.demo-instance[1]: Still creating... [00m30s elapsed]
aws_instance.demo-instance[1]: Still creating... [00m40s elapsed]
aws_instance.demo-instance[1]: Still creating... [00m50s elapsed]
aws_instance.demo-instance[1]: Creation complete after 53s [id=i-02fa2612fd694c6
6d]

Apply complete! Resources: 1 added, 1 changed, 0 destroyed.

Outputs:

ec2_public_dns = [
  "ec2-16-145-86-114.us-west-2.compute.amazonaws.com",
  "ec2-44-255-211-249.us-west-2.compute.amazonaws.com",
]
ec2_public_ips = [
  "16.145.86.114",
  "44.255.211.249",
]
molly@mollys-MacBook-Pro hw2 %
```

Instances (2) Info

Connect | Instance state | Actions | Launch instances

Find Instance by attribute or tag (case-sensitive) | All states

Instance state = running | Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
hw2-ec2-instance-0	i-03be1d2eec2d10b38	Running	t2.micro	2/2 checks passed	View alarms +	us-west-2a	ec2-16-145-86-114.us-...	16.145.86.114	–
hw2-ec2-instance-1	i-02fa2612fd694c66d	Running	t2.micro	2/2 checks passed	View alarms +	us-west-2a	ec2-44-255-211-249.us...	44.255.211.249	–

Final Result:

Get request successfully running in two different instances:

```
hw2 — ec2-user@ip-172-31-38-98:~ — ssh -i YumengZeng-cs6650-hw1...
        "artist": "Sarah Vaughan",
        "price": 39.99
    }
][ec2-user@ip-172-31-38-98 ~]$ curl http://16.145.86.114:8080/albums
[
    {
        "id": "1",
        "title": "Blue Train",
        "artist": "John Coltrane",
        "price": 56.99
    },
    {
        "id": "2",
        "title": "Jeru",
        "artist": "Gerry Mulligan",
        "price": 17.99
    },
    {
        "id": "3",
        "title": "Sarah Vaughan and Clifford Brown",
        "artist": "Sarah Vaughan",
        "price": 39.99
    }
][ec2-user@ip-172-31-38-98 ~]$
```

```
hw2 — ec2-user@ip-172-31-35-221:~/cs6650-distributed-systems/hw2...
[ec2-user@ip-172-31-35-221 hw2]$ docker run -d -p 8080:8080 --name my-running-ap
p my-go-app
f53ca9d71f4acf04c984df77be1b8e78b7459a2be0519ae879143dff3e2e4e8c
[ec2-user@ip-172-31-35-221 hw2]$ curl http://44.255.211.249:8080/albums
[
    {
        "id": "1",
        "title": "Blue Train",
        "artist": "John Coltrane",
        "price": 56.99
    },
    {
        "id": "2",
        "title": "Jeru",
        "artist": "Gerry Mulligan",
        "price": 17.99
    },
    {
        "id": "3",
        "title": "Sarah Vaughan and Clifford Brown",
        "artist": "Sarah Vaughan",
        "price": 39.99
    }
][ec2-user@ip-172-31-35-221 hw2]$
```

Inconsistency

```
molly@mollys-MacBook-Pro hw2 % python3 test_consistency.py

uhoh... what happened?
molly@mollys-MacBook-Pro hw2 % python3 test_consistency.py

/Users/molly/Library/Python/3.9/lib/python/site-packages/urllib3/__init__.py:35: NotOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.1.1+, currently the 'ssl' module is compiled wi
ibreSSL 2.8.3'. See: https://github.com/urllib3/urllib3/issues/3020
  warnings.warn(
Starting data test...
Instance 1 response: [
    {
        "id": "1",
        "title": "Blue Train",
        "artist": "John Coltrane",
        "price": 56.99
    },
    {
        "id": "2",
        "title": "Jeru",
        "artist": "Gerry Mulligan",
        "price": 17.99
    },
    {
        "id": "3",
        "title": "Sarah Vaughan and Clifford Brown",
        "artist": "Sarah Vaughan",
        "price": 39.99
    }
]


and...
Instance 2 response: [
    {
        "id": "1",
        "title": "Blue Train",
        "artist": "John Coltrane",
        "price": 56.99
    },
    {
        "id": "2",
        "title": "Jeru",
        "artist": "Gerry Mulligan",
        "price": 17.99
    },
    {
        "id": "3",
        "title": "Sarah Vaughan and Clifford Brown",
        "artist": "Sarah Vaughan",
        "price": 39.99
    }
]
```

```
and adding...
Instance 2 response: {
    "id": "4",
    "title": "The Modern Sound of Betty Carter",
    "artist": "Betty Carter",
    "price": 49.99
}
Instance 1 response: [
    {
        "id": "1",
        "title": "Blue Train",
        "artist": "John Coltrane",
        "price": 56.99
    },
    {
        "id": "2",
        "title": "Jeru",
        "artist": "Gerry Mulligan",
        "price": 17.99
    },
    {
        "id": "3",
        "title": "Sarah Vaughan and Clifford Brown",
        "artist": "Sarah Vaughan",
        "price": 39.99
    }
]


and...
Instance 2 response: [
    {
        "id": "1",
        "title": "Blue Train",
        "artist": "John Coltrane",
        "price": 56.99
    },
    {
        "id": "2",
        "title": "Jeru",
        "artist": "Gerry Mulligan",
        "price": 17.99
    },
    {
        "id": "3",
        "title": "Sarah Vaughan and Clifford Brown",
        "artist": "Sarah Vaughan",
        "price": 39.99
    },
    {
        "id": "4",
        "title": "The Modern Sound of Betty Carter",
        "artist": "Betty Carter",
        "price": 49.99
    }
```

Observation:

In Part 4, I successfully deployed two instances using Terraform. By running the Python test script, I observed that data added to Instance 2 was not reflected in Instance 1. This demonstrated that in-memory storage is insufficient for distributed systems, as it leads to data inconsistency across multiple nodes. A shared database is required for a truly scalable architecture.